Input to Quadratic Equation Solver

Put quadratic equation in the form:

$$\frac{1}{2}[X^T P X] + q^T X$$

$$Gx \leq h$$

where:

X *is a Vector of size* $d*1$,

P *is a Matrix of size* $d*d$,

$\mathbf{q^T}$ *is a Vector of size* $1*d$,

?

$$Ax = b$$

Put all ≤ , ≥ constraints in the form:

where:

m *is Number of constraints* ,

X *is a Vector of size* $d*1$,

G *is a Matrix of size* $m*d$,

h *is a Vector of size* $m*1$,

Put all *Equal "="* constraints in the form:

where:

m *is Number of constraints* ,

X *is a Vector of size* $d*1$,

A *is a Matrix of size* $m*d$,

b *is a Vector of size* $m*1$,